

Hongxin Zhang

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EDUCATION

University of Massachusetts Amherst, Amherst, MA, USA

Ph.D. in Computer Science

Sep. 2023 - Present

Advisor: Chuang Gan

University of Massachusetts Amherst, Amherst, MA, USA

Master of Science in Computer Science

Sep. 2023 - May 2025

Advisor: Chuang Gan

Shanghai Jiao Tong University, Shanghai, China

B.E. in Computer Science, ACM Honors Class

Sep. 2019 - June 2023

Advisor: Yong Yu, Diyi Yang

PUBLICATIONS

1. Xiangye Lin*, **Hongxin Zhang***, Ruxi Deng, Qinhong Zhou, Chuang Gan. "Sentinel: Embodied Cooperative Spatial Reasoning and Planning" *European Conference on Computer Vision (ECCV)*, 2026
2. Qinhong Zhou*, **Hongxin Zhang***, Xiangye Lin*, Zheyuan Zhang*, Yutian Chen, Wenjun Liu, Zunzhe Zhang, Sunli Chen, Lixing Fang, Qiushi Lyu, Xinyu Sun, Jincheng Yang, Zeyuan Wang, Bao Chi Dang, Zhehuan Chen, Daksha Ladia, Jiageng Liu, Chuang Gan. "Virtual Community: An Open World for Humans, Robots, and Society" *International Conference on Learning Representations (ICLR)*, 2026.
3. **Hongxin Zhang***, Zheyuan Zhang*, Zeyuan Wang*, Zunzhe Zhang, Lixing Fang, Qinhong Zhou, Chuang Gan. "Ella: Embodied Social Agents with Lifelong Memory" *International Conference on Machine Learning (ICML) MAS Workshop*, 2025
4. Haoyu Zhen*, Qiao Sun*, **Hongxin Zhang**, Junyan Li, Siyuan Zhou, Yilun Du, Chuang Gan. "TesserAct: Learning 4D Embodied World Models" *International Conference on Computer Vision (ICCV)*, 2025
5. Yuncong Yang*, Han Yang*, Jiachen Zhou, Peihao Chen, **Hongxin Zhang**, Yilun Du, Chuang Gan. "3D-Mem: 3D Scene Memory for Embodied Exploration and Reasoning" *The IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2025
6. **Hongxin Zhang***, Zeyuan Wang*, Qiushi Lyu*, Zheyuan Zhang, Sunli Chen, Tianmin Shu, Yilun Du, Behzad Dariush, Kwonjoon Lee, Chuang Gan. "COMBO: Compositional World Models for Embodied Multi-Agent Cooperation" *International Conference on Learning Representations (ICLR)*, 2025.
7. Weihua Du*, Qiushi Lyu*, Jiaming Shan, Zhenting Qi, **Hongxin Zhang**, Sunli Chen, Andi Peng, Tianmin Shu, Kwonjoon Lee, Behzad Dariush, Chuang Gan. "Constrained Human-AI Cooperation: An Inclusive Embodied Social Intelligence Challenge" *Neural Information Processing Systems (NeurIPS)*, 2024.
8. **Hongxin Zhang***, Weihua Du*, Jiaming Shan, Qinhong Zhou, Yilun Du, Joshua B. Tenenbaum, Tianmin Shu, Chuang Gan. "Building Cooperative Embodied Agents Modularly with Large Language Models" *International Conference on Learning Representations (ICLR)*, 2024.
9. Qinhong Zhou, Sunli Chen, Yisong Wang, Haozhe Xu, Weihua Du, **Hongxin Zhang**, Yilun Du, Joshua B. Tenenbaum, Chuang Gan. "HAZARD Challenge: Embodied Decision Making in Dynamically Changing Environments" *International Conference on Learning Representations (ICLR)*, 2024.
10. Zhiqing Sun, Yikang Shen, **Hongxin Zhang**, Qinhong Zhou, Zhenfang Chen, David Cox, Yiming Yang, Chuang Gan. "SALMON: Self-Alignment with Principle-Following Reward Models" *International Conference on Learning Representations (ICLR)*, 2024.
11. Zhiqing Sun, Yikang Shen, Qinhong Zhou, **Hongxin Zhang**, Zhenfang Chen, David Cox, Yiming Yang, Chuang Gan. "Principle-Driven Self-Alignment of Language Models from Scratch with Minimal Human Supervision" *Neural Information Processing Systems (NeurIPS)*, 2023.
12. Omar Shaikh, **Hongxin Zhang**, William Held, Michael Bernstein, Diyi Yang. "On Second Thought, Let's Not Think Step by Step! Bias and Toxicity in Zero-Shot Reasoning" *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (ACL)*, 2023.
13. Bolin Lai*, **Hongxin Zhang***, Miao Liu*, Aryan Pariani*, Fiona Ryan, Wenqi Jia, Shirley Anugrah Hayati, James M. Rehg, Diyi Yang. "Werewolf Among Us: A Multimodal Dataset for Modeling Persuasion Behaviors in Social Deduction Games" *Findings of the Association for Computational Linguistics: ACL*, 2023.
14. Albert Lu*, **Hongxin Zhang***, Yanzhe Zhang, Xuezhi Wang, Diyi Yang. "Bounding the Capabilities of Large Language Models in Open Text Generation with Prompt Constraints" *Findings of the Association for Computational Linguistics: EACL*, 2023.
15. **Hongxin Zhang**, Yanzhe Zhang, Ruiyi Zhang, Diyi Yang. "Robustness of Demonstration-based Learning Under Limited Data Scenario" *Empirical Methods in Natural Language Processing (EMNLP)*, 2022.

PREPRINTS

1. **Hongxin Zhang**, Chunru Lin, Junyan Li, Zhou Xian, Tsun-Hsuan Wang, Chuang Gan. "GS-Agent: Creating 4D Physical Worlds With Generative Simulation"

2. Chunru Lin*, **Hongxin Zhang***, Fenghao Yu, Zhehuan Chen, Thomas L. Griffiths, Yejin Choi, David Held, Chuang Gan. “RoboW-its: Unexpected Challenges for Robotic Creative Problem Solving”

RESEARCH EXPERIENCE

University of Massachusetts Amherst

Amherst, Massachusetts

Graduate Research Assistant, advised by Prof. Chuang Gan

Sep. 2023 - Present

- **COMBO: Compositional World Models for Embodied Multi-Agent Cooperation**

Hongxin Zhang*, Zeyuan Wang*, Qiushi Lyu*, Zheyuan Zhang, Sunli Chen, Tianmin Shu, Yilun Du, Behzad Dariush, Kwonjoon Lee, Chuang Gan

ICLR 2025 Poster

- Proposed to learn a compositional world model for multi-agent cooperation by factorizing joint actions of an arbitrary number of agents and compositionally generating the video to enable accurate simulation of multiple sets of actions on the world state.
- Introduced a Compositional World Model-based embodied multi-agent planning framework to empower the agents to imagine how different actions may affect the world with other agents in the long run and plan more cooperatively.
- Trained a 200M video diffusion model from scratch, another two smaller diffusion models for inpainting and super-resolution, and LoRA fine-tuned LLaVA-7B as planning sub-modules.

- **Building Cooperative Embodied Agents Modularly with Large Language Models**

Hongxin Zhang*, Weihua Du*, Jiaming Shan, Qinhong Zhou, Yilun Du, Joshua B. Tenenbaum, Tianmin Shu, Chuang Gan

ICLR 2024 Poster

- Conducted the first systematic study on LLMs’ capacity for planning and communication in embodied multi-agent cooperation.
- Introduced a novel framework that utilizes LLMs to build cooperative embodied agents, integrating perception, memory, planning, communication, and motion, surpassing strong planning-based methods.
- Conducted a user study to evaluate the possibility of achieving effective and trustworthy human-AI cooperation using LLMs.

- **SALMON: Self-Alignment with Principle-Following Reward Models**

Zhiqing Sun, Yikang Shen, **Hongxin Zhang**, Qinhong Zhou, Zhenfang Chen, David Cox, Yiming Yang, Chuang Gan

ICLR 2024 Poster

- Proposed a new RLAIFF paradigm for self-aligning language models from scratch, using only a small set of human-defined principles as guidance.
- Open-sourced Large Language Model Dromedary-2, which is trained with the SALMON paradigm on the LLaMA-2-70b base language model, with Principle-Driven Self-Alignment as the Supervised Fine-Tuning (SFT) strategy to initialize the policy model.

- **Principle-Driven Self-Alignment of Language Models from Scratch with Minimal Human Supervision**

Zhiqing Sun, Yikang Shen, Qinhong Zhou, **Hongxin Zhang**, Zhenfang Chen, David Cox, Yiming Yang, Chuang Gan

Neurips 2023 Spotlight

- Proposed a novel approach called SELF-ALIGN, which combines principle-driven reasoning and the generative power of LLMs for the self-alignment of the AI agents with minimal human supervision.
- Released Dromedary, an open-source self-aligned language model trained with minimal human supervision.

Stanford University

Stanford, California

Visiting Student Researcher, advised by Prof. Diyi Yang

Sep. 2022 - Jan. 2023

- **On Second Thought, Let’s Not Think Step by Step! Bias and Toxicity in Zero-Shot Reasoning**

Omar Shaikh, **Hongxin Zhang**, William Held, Michael Bernstein, Diyi Yang **ACL 2023**

- Performed a controlled evaluation of zero-shot CoT across two sensitive domains: harmful questions & stereotype benchmarks.
- Found that using zero-shot CoT reasoning in a prompt can significantly increase a model’s likelihood of producing undesirable output.

- **Werewolf Among Us: A Multimodal Dataset for Modeling Persuasion Behaviors in Social Deduction Games**

Bolin Lai*, **Hongxin Zhang***, Miao Liu*, Aryan Pariani*, Fiona Ryan, Wenqi Jia, Shirley Anugrah Hayati, James M. Rehg, Diyi Yang. **ACL 2023**

- Presented the first multimodal dataset for persuasion modeling which is collected in naturalistic social game scenarios with intensive in-person conversations with multiple players.
- Conducted comprehensive experiments to show the importance of context and visual signals for persuasion strategy prediction.
- Provided additional experiment results to investigate model generalization on the persuasion modeling task, and discuss how persuasion strategy influences the game voting outcome.

- **Bounding the Capabilities of Large Language Models in Open Text Generation with Prompt Constraints**

Albert Lu*, **Hongxin Zhang***, Yanzhe Zhang, Xuezhi Wang, Diyi Yang. **EACL 2023**

- Provided a taxonomy of prompts containing stylistic or structural constraints to facilitate finer-grained analyses of open text generation.
- Conducted a systematic experiment using proposed taxonomy by creating 288 different prompts and evaluating 3000+ generated outputs to analyze the capabilities and limitations of current LLMs on open-ended text generation.
- Analyzed in-context mitigation strategies to improve model performance.

Georgia Institute of Technology

Visiting Student Researcher, advised by Prof. Diyi Yang

Atlanta, Georgia
Aug. 2021 - Aug. 2022

- **Robustness of Demonstration-based Learning Under Limited Data Scenario**

Hongxin Zhang, Yanzhe Zhang, Ruiyi Zhang, Diyi Yang. **EMNLP 2022**

- Investigated the robustness of demonstration-based learning by designing pathological demonstrations.
- Experiments show demonstrations composed of random tokens still make the model a better few-shot learner; the length of random demonstrations and the relevance of random tokens are the main factors affecting the performance; demonstrations increase the confidence of model predictions on captured superficial patterns.

SELECTED PROJECTS

Mx* Compiler

Java

- Developed a compiler that compiles C-and-Java-like language (Mx*) to Assembly Language
- Designed LLVM-like IR
- Implemented optimizations based on data-flow and control-flow analysis

RISC-V CPU

Verilog

- Designed a RISC-V CPU that supports RV32I Base Integer Instruction Set V2.0 (2.1 2.6) with the following property:
 - 5-stage pipelined
 - 1KB iCache, direct mapped
 - Branch Prediction using 2-bit saturating counter BHT with Branch Target Buffer (Size is 128*4 Byte)
 - Running on FPGA with 200MHz
 - Data forwarding supported

TEACHING EXPERIENCE

Teaching Assistant: Great Ideas in Computer Science

Fall 2020

Teaching Assistant: Data Structure

Spring 2021

Student Instructor: Principle and Practice of Computer Algorithms

Summer 2021

Teaching Assistant: Mathematical Logic

Fall 2021

Teaching Assistant: Machine Learning

Spring 2022

Teaching Assistant: CICS-160 Object Oriented Programming

Spring 2024

Teaching Assistant: COMPSCI-603 Robotics

Spring 2025

SKILLS AND INTERESTS

Programming: Python / C++ / Java / Go / Verilog

Language: Mandarin (native), English (TOEFL 109/120), Latin (Beginner)

Interests: Literature, Movie, Billiards

HONORS AND AWARDS

Shanghai Excellent Graduate (Awarded for overall performance in undergraduate career)

2023

Zhiyuan Outstanding Student Scholarship (Highest award for undergraduate in SJTU)

2023

Foresight-Sequoia Scholarship (5 winners at Zhiyuan College)

2022

Academic Excellence Scholarship

2020, 2021, 2022

35th China's National Olympiad in Informatics (NOI) Silver Medal

2018